

ASSIGNMENT 1

WEB DEVELOPMENT

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1. Explain the difference between frontend, backend, and full-stack development with suitable real-world examples.

Frontend Development: Frontend is the part of a website that users can see and interact with. It includes the design, buttons, menus, forms, images, etc. Common technologies include HTML, CSS and JavaScript. Example: The Instagram login page that a user sees and interacts with.

Backend Development: Backend is the server-side part of a website. It handles business logic, user authentication, data processing and communication with databases. Common technologies include Java, Python, Node.js and PHP. Example: When you log in to Instagram, the backend checks your username and password.

Full-Stack Development: A full-stack developer works on both frontend and backend and understands how they work together. Example: A developer who creates an Amazon product page (frontend) and also develops the server and database that handle orders and payments (backend).

Frontend	Backend	Full-Stack
User interface	Server-side logic	Frontend + Backend
HTML, CSS, JS	Java, Python, Node.js	Works with both
What user sees	Behind-the-scenes processing	Complete web application

2. Create a simple diagram showing how the client-server model works in web architecture.

Client-Server Model: The client, usually a web browser, sends a request to a server. The server processes the request and sends a response back to the client.



Example: When we open a website, the browser sends a request to the website's server. The server processes the request and sends the webpage back to the browser.

3. Describe how a browser requests and displays a web page from a web server.

- The user enters a website address in the browser.
- The browser finds the IP address of the web server using DNS.
- The browser sends an HTTP/HTTPS request to the server.
- The web server processes the request.
- The server sends files such as HTML, CSS, JavaScript and images.
- The browser receives these files.
- The browser parses the HTML and applies CSS and JavaScript.
- Finally, the browser renders and displays the webpage.

4. Identify and list the tools required to set up a web development environment. Explain the purpose of each.

Tool	Purpose
VS Code	Used to write and edit HTML, CSS and JavaScript code.
Web Browser	Used to run and test websites. Examples: Chrome, Firefox and Edge.
HTML	Creates the structure of a webpage.
CSS	Adds styling and design to a webpage.
JavaScript	Adds functionality and interactivity.
Git	Used for version control and tracking changes in code.
Node.js	Provides a JavaScript runtime for server-side development and tools.
Live Server	Helps preview webpages and automatically refreshes them when code changes.

5. Explain what a web server is and give examples of commonly used servers.

A web server is a computer or software that receives requests from clients such as web browsers and sends web resources such as HTML, CSS, JavaScript, images or other data in response.

Examples of commonly used web servers: Apache HTTP Server, Nginx, Microsoft IIS, LiteSpeed and Apache Tomcat.

Example: When a user visits a website, the browser sends a request to the web server. The web server returns the required webpage to the browser.

6. Define the roles of a frontend developer, backend developer, and database administrator in a project.

Frontend Developer: Creates the user interface, uses HTML/CSS/JavaScript, makes websites responsive and user-friendly, and handles buttons, menus and forms.

Backend Developer: Develops server-side logic, creates APIs, handles authentication and business logic, and connects the application to the database.

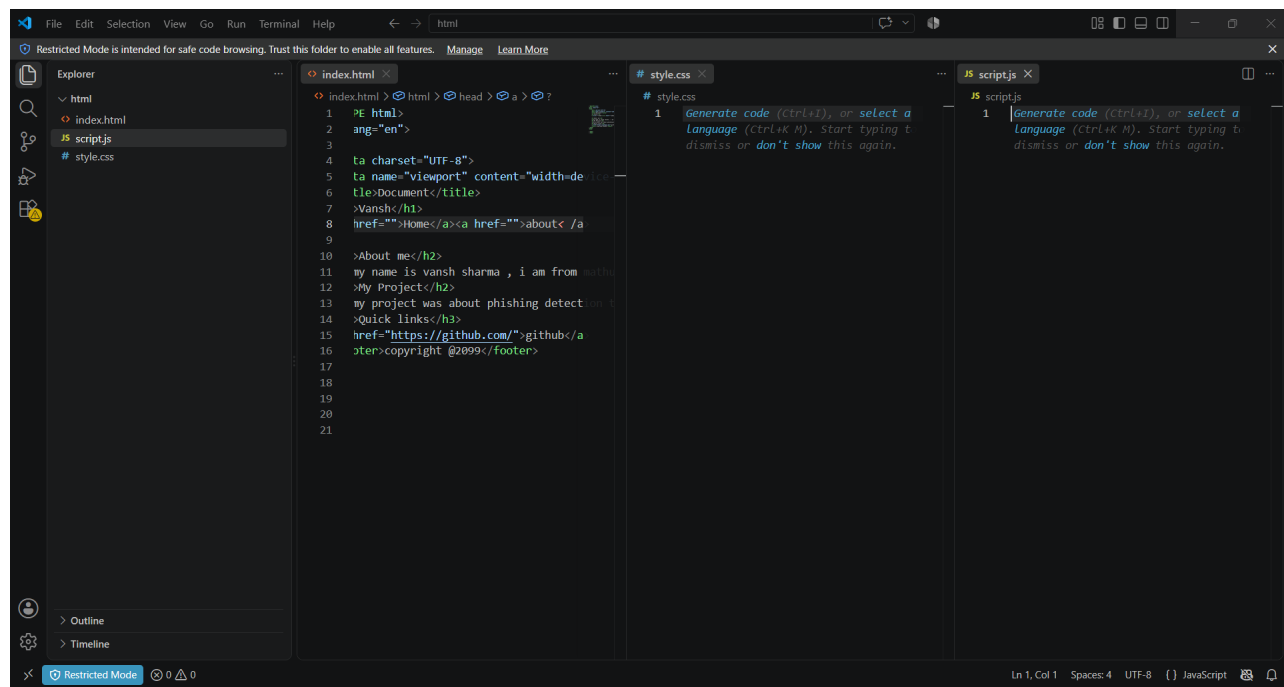
Database Administrator (DBA): Manages databases, creates and maintains database tables, manages data security and backups, and monitors database performance.

7. Install VS Code and configure it for HTML, CSS, and JavaScript development. Take a screenshot of the setup.

VS Code was configured for HTML development and the webpage was successfully run using Live Server. The following screenshots show the development environment and the resulting webpage.

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Webpage running successfully through Live Server

8. Explain the difference between static and dynamic websites. Provide an example of each.

Static Website: A static website displays fixed content. The content generally remains the same for every user unless the developer manually changes the files. Example: A simple portfolio or personal information website made with HTML and CSS.

Dynamic Website: A dynamic website can generate different content based on users, database information or user actions. Example: Amazon, Instagram, YouTube and online banking websites.

Static Website	Dynamic Website
Fixed content	Content can change
Usually simpler	More complex
HTML/CSS commonly used	Frontend + backend + database

Database is not necessarily required	Often uses a database
Example: Portfolio	Example: Amazon

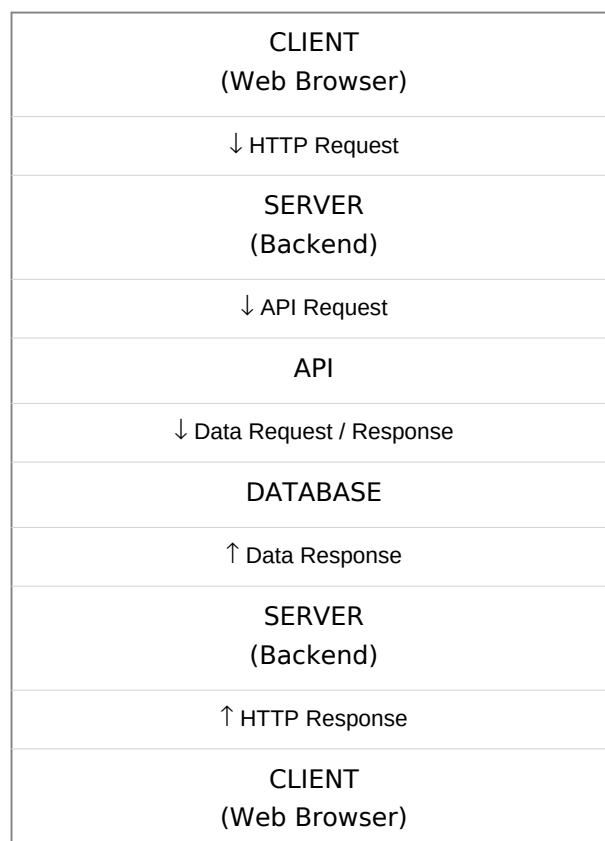
9. Research and list five web browsers. Explain how rendering engines differ between them.

A rendering engine is responsible for processing HTML and CSS and displaying the webpage on the screen.

Browser	Rendering Engine
Google Chrome	Blink
Microsoft Edge	Blink
Mozilla Firefox	Gecko
Apple Safari	WebKit
Opera	Blink

Different browsers may use different rendering engines. These engines interpret HTML, CSS and other web technologies and convert them into the visual webpage. Because engines have different implementations and update cycles, a website may sometimes look or behave slightly differently across browsers. For example, Chrome and Edge both use Blink, while Firefox uses Gecko and Safari uses WebKit.

10. Draw a labeled diagram showing the basic web architecture flow — client, server, database, and APIs.



The client is usually a web browser. It sends a request to the server. The backend server processes the request and may communicate with the database through APIs. The database stores and retrieves information. The server then sends the required response back to the client, where the information is displayed to the user.